

University of Moratuwa

Department of Electronic and Telecommunication Engineering

EN3330

Assignment 03

Lagrangian Duality, Saddle Points
and Weak Duality

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Source code: `https://github.com/rajinthanr/en3330-lagrangian-duality`

Overview

All three questions are based on the same problem,

$$\min_x \; f(x) = x^2 \quad \text{subject to} \quad h(x) = 1 - x \le 0, \qquad \mathcal{L}(\mu, x) = x^2 + \mu(1 - x), \quad \mu \ge 0. \tag{1}$$

The answer to the primal problem is clear. Since x^2 is increasing on $[1, \infty)$, the smallest value is at $x^* = 1$, and $p^* = f(x^*) = 1$. In this report, the same problem is studied using the Lagrangian.

All the figures were drawn using one Python script (NumPy and Matplotlib). The same script also checks every analytical result numerically, using grid search. The results of these checks are given in the relevant sections, and the full output is given in Appendix A.

Source code: `https://github.com/rajinthanr/en3330-lagrangian-duality`
File `src/duality_simulation.py`. It can be run using `python3 src/duality_simulation.py`. The full code is given in Appendix B.

1 The Lagrangian, the dual function and the primal function

1.1 (a) $\mathcal{L}(\mu, x)$ as a function of x

When μ is fixed, $\mathcal{L}(\mu, \cdot)$ is a convex parabola in x. Its minimum point is found by setting the derivative to zero:

$$\frac{\partial \mathcal{L}}{\partial x} = 2x - \mu = 0 \implies x(\mu) = \arg\min_x \mathcal{L}(\mu, x) = \frac{\mu}{2}, \qquad \mathcal{L}\big(\mu, x(\mu)\big) = \mu - \frac{\mu^2}{4}. \tag{2}$$

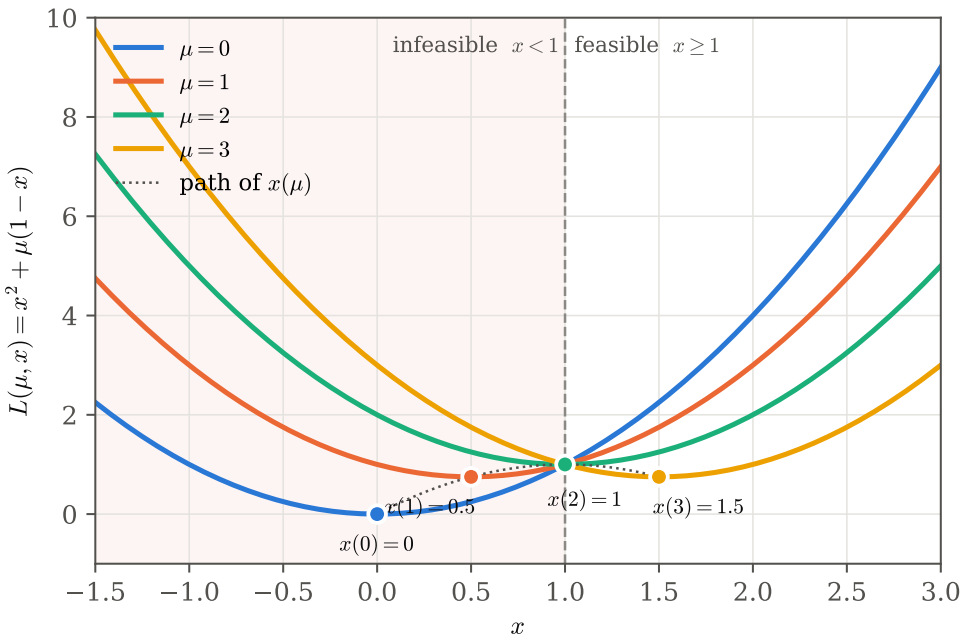


Figure 1: $\mathcal{L}(\mu, x) = x^2 + \mu(1 - x)$ for $\mu = 0, 1, 2, 3$. The dots show the minimum points $x(\mu) = \mu/2$. The shaded area is not feasible ($x < 1$). The dotted curve shows how the minimum point moves when μ changes.

How the minimum point moves. When μ increases, the minimum point $x(\mu) = \mu/2$ moves to the right. It moves by $1/2$ for every unit increase in μ (Figure 1). The term $\mu(1 - x)$ adds a cost to the points with $x < 1$ and reduces the cost of the points with $x > 1$. Because of this, the lowest point of the parabola moves towards the feasible region.

Table 1: Minimum points of $\mathcal{L}(\mu, \cdot)$. The values from the formula are the same as the values from a grid search over $x \in [-10, 10]$ with step 5×10^{-5}.

μ	$x(\mu) = \mu/2$	$\min_x \mathcal{L}(\mu, x)$	Feasible?
0	0.0	0.00	no
1	0.5	0.75	no
2	1.0	1.00	yes (on the boundary)
3	1.5	0.75	yes (inside)

- When $\mu < 2$, the minimum point is still not feasible ($x(\mu) < 1$). The value of μ is not large enough.
- When $\mu = 2$, the minimum point is exactly on the boundary, $x(2) = 1 = x^*$.
- When $\mu > 2$, the minimum point goes inside the feasible region. It is feasible, but it is not the optimal point. The value of μ is too large.

It can also be seen that all the curves go through the point $(1, 1)$. The reason is that $h(1) = 0$, so $\mathcal{L}(\mu, 1) = 1$ for every μ. On the boundary, the value of μ does not change the Lagrangian.

1.2 (b) The dual function

By substituting $x(\mu)$ from (2) into the Lagrangian,

$$d(\mu) = \min_x \mathcal{L}(\mu, x) = \left(\frac{\mu}{2}\right)^2 + \mu\left(1 - \frac{\mu}{2}\right) = \mu - \frac{\mu^2}{4}, \qquad \mu \geq 0. \tag{3}$$

Since $d''(\mu) = -\frac{1}{2} < 0$, the dual function is a concave parabola. To find its maximum,

$$d'(\mu) = 1 - \frac{\mu}{2} = 0 \implies \boxed{\mu^* = 2, \qquad d^* = d(2) = 1.} \tag{4}$$

This value satisfies $\mu^* \geq 0$, so the condition $\mu \geq 0$ does not affect the answer. A grid search over $\mu \in [0, 10]$ gives the same values (Appendix A).

Discussion. For every μ, $d(\mu) \leq 1$, and the maximum value is exactly $p^* = 1$. Therefore the *duality gap is zero*. This is the expected result. The problem is convex (f is convex and h is linear), and there is a point which satisfies the constraint strictly (for example $x = 2$ gives $h(2) < 0$). So Slater's condition is satisfied, and strong duality holds. It can also be seen that $\mu^* = 2$ is the same value of μ for which the minimum point in part (a) is at $x^* = 1$. That is, $x(\mu^*) = x^*$.

The value of μ^* also has a practical meaning. Suppose the constraint is changed to $x \geq b$ with $b \geq 0$. Then the optimal value is $p^*(b) = b^2$, and $\mathrm{d}p^*/\mathrm{d}b = 2b = 2 = \mu^*$ at $b = 1$. This means that if the constraint is made tighter by a small amount δ, the optimal cost increases by about 2δ. For this reason, μ^* is called the "shadow price" of the constraint.

1.3 (c) The primal function

When x is fixed, $\mathcal{L}(\mu, x) = x^2 + \mu(1 - x)$ is a straight line in μ with slope $1 - x = h(x)$. Therefore, the maximum over $\mu \geq 0$ is

$$g(x) = \max_{\mu \geq 0} \mathcal{L}(\mu, x) = x^2 + \sup_{\mu \geq 0} \mu(1 - x) = \begin{cases} x^2, & x \geq 1 \quad (\text{slope} \leq 0,\ \text{best } \mu = 0), \\ +\infty, & x < 1 \quad (\text{slope} > 0,\ \mu \to \infty). \end{cases} \tag{5}$$

To show this using a graph, Figure 3 plots $g_M(x) = \max_{0 \leq \mu \leq M} \mathcal{L}(\mu, x)$ for larger and larger values of M.

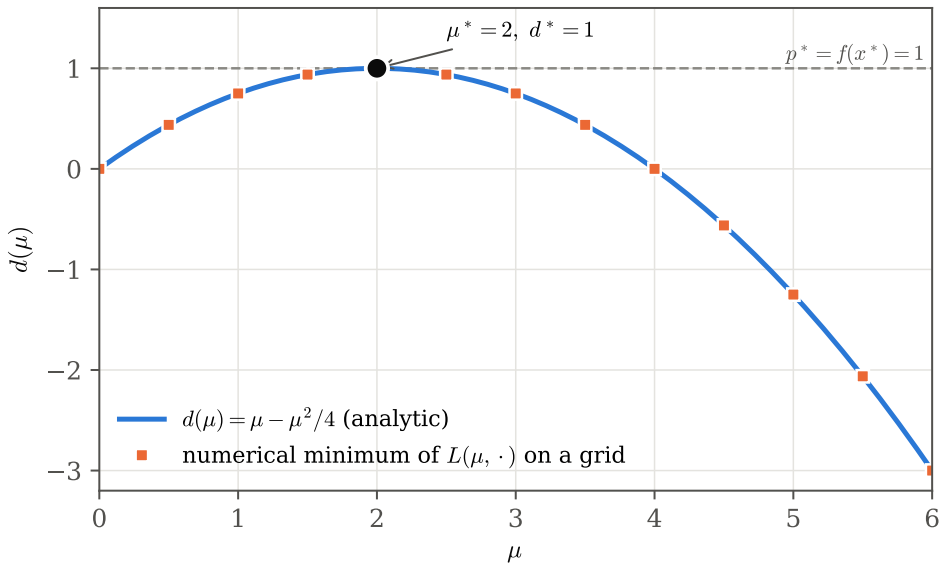


Figure 2: Dual function $d(\mu) = \mu - \mu^2/4$ (line) and the values from the numerical minimisation of $\mathcal{L}(\mu, \cdot)$ (squares). The maximum value $d^* = 1$ at $\mu^* = 2$ is equal to the primal optimal value $p^* = 1$ (dashed line).

- In the feasible region ($x \geq 1$), all the curves are the same as x^2 for any value of M.
- In the infeasible region ($x < 1$), $g_M(x) = x^2 + M(1-x)$ becomes larger and larger when M increases. It goes to $+\infty$ when $M \to \infty$.

The values in Table 2 show the same result. At $x = 0$ the value is equal to M. Even at $x = 0.99$, the value keeps increasing with M.

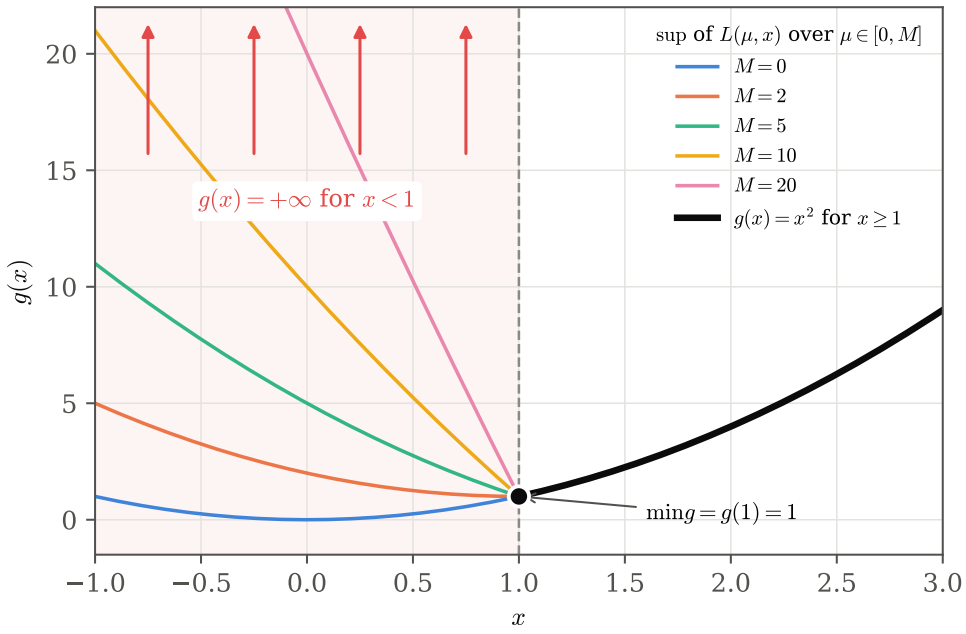


Figure 3: Primal function. The coloured curves are $\max_{0 \leq \mu \leq M} \mathcal{L}(\mu, x)$ for different values of M. They are equal to x^2 when $x \geq 1$ and go to infinity when $x < 1$. Therefore $g(x) = x^2 + I_{\{x \geq 1\}}(x)$. Its minimum is at $x = 1$, where $g(1) = 1$.

Why $\min_x g(x)$ is the same as problem (1). From (5), $g = f + I_{\mathcal{F}}$. Here $I_{\mathcal{F}}$ is 0 on the feasible set

Table 2: Values of $\max_{0\le\mu\le M} \mathcal{L}(\mu, x)$ from the simulation.

x	$M = 1$	$M = 10$	$M = 100$	$M = 1000$	x^2
−1.00	3.00	21.00	201.00	2001.00	1.00
0.00	1.00	10.00	100.00	1000.00	0.00
0.99	0.99	1.08	1.98	10.98	0.98
1.00	1.00	1.00	1.00	1.00	1.00
2.00	4.00	4.00	4.00	4.00	4.00

$\mathcal{F} = \{x \ge 1\}$ and $+\infty$ outside it. Every infeasible point has an infinite value, so it cannot be the minimum of g. Inside $\mathcal{F}$, g is the same as the original objective. Therefore,

$$\min_x g(x) = \min_{x\ge1} x^2 = p^* = 1, \qquad \arg\min_x g = x^* = 1.$$

Both problems have the same minimum point and the same minimum value. In other words, the constrained problem can be written as a problem without constraints, $p^* = \min_x \max_{\mu\ge0} \mathcal{L}(\mu, x)$. The dual problem in part (b) is the same expression with the order changed, $d^* = \max_{\mu\ge0} \min_x \mathcal{L}(\mu, x)$. For this problem, both orders give the same value, $d^* = p^* = 1$.

2 Saddle point of the Lagrangian

In this question, $x^* = 1$ and $\mu^* = 2$ from Question 1 are used.

2.1 (a) (x^*, μ^*) is a saddle point

(i) $\mathcal{L}(\mu^*, x^*) = 1^2 + 2(1 - 1) = \boxed{1}$.

(ii) $\mathcal{L}(\mu, x^*) = 1^2 + \mu(1 - 1) = 1$ for every $\mu \ge 0$. This does not depend on μ, because the constraint is active at x^* ($h(x^*) = 0$). So the term with μ becomes zero. Therefore,

$$\mathcal{L}(\mu, x^*) = 1 \le 1 = \mathcal{L}(\mu^*, x^*) \qquad \forall\, \mu \ge 0,$$

and the left inequality is satisfied (with equality).

(iii) $\mathcal{L}(\mu^*, x) = x^2 + 2(1 - x) = x^2 - 2x + 2 = (x - 1)^2 + 1$. Since $(x - 1)^2 \ge 0$,

$$\mathcal{L}(\mu^*, x) = (x - 1)^2 + 1 \ge 1 = \mathcal{L}(\mu^*, x^*) \qquad \forall\, x.$$

Equality is only at $x = x^* = 1$. Therefore the right inequality is also satisfied.

(iv) The two graphs are shown in Figure 4. Both graphs go through the common value $\mathcal{L}(\mu^*, x^*) = 1$. In the μ direction, the Lagrangian is constant, so μ^* is a maximum point (but not the only one). In the x direction, it is a parabola with its minimum at x^*.

Figure 5 shows the whole function as a contour plot. The contour $\mathcal{L} = 1$ is made of two lines, $x = 1$ and $x = \mu - 1$. These two lines cross at $(\mu^*, x^*) = (2, 1)$, which is the usual shape near a saddle point. Both inequalities were also checked numerically for $\mu \in [0, 50]$ and $x \in [-20, 20]$. The results were $\max_\mu \mathcal{L}(\mu, x^*) = 1$ and $\min_x \mathcal{L}(\mu^*, x) = 1$. The largest difference between $\mathcal{L}(2, x)$ and $(x - 1)^2 + 1$ was about 10^{-13}, which is only a rounding error.

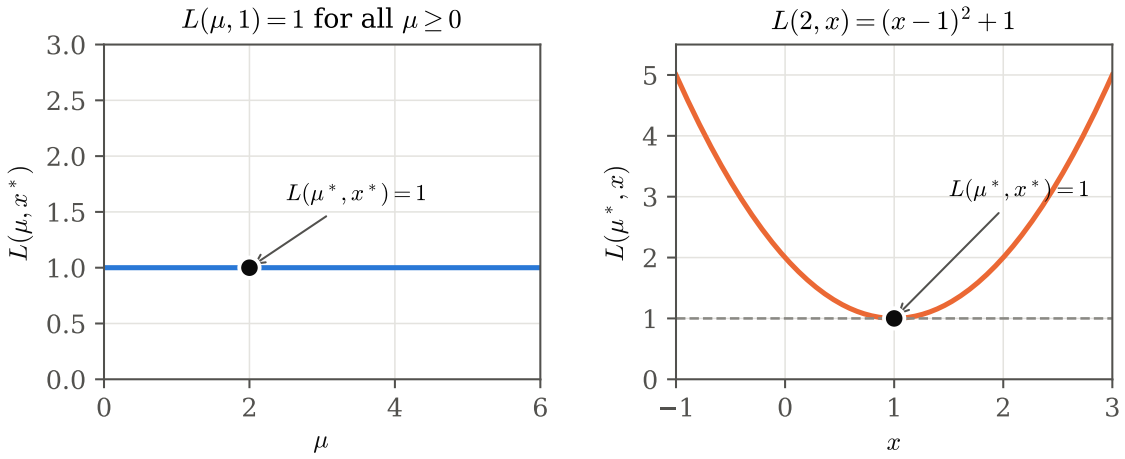


Figure 4: $\mathcal{L}$ through the saddle point. Left: $\mathcal{L}(\mu, x^*)$ is always equal to 1. Right: $\mathcal{L}(\mu^*, x) = (x-1)^2 + 1$ has its minimum value of 1 at $x^* = 1$. The black dot shows the common value $\mathcal{L}(\mu^*, x^*) = 1$.

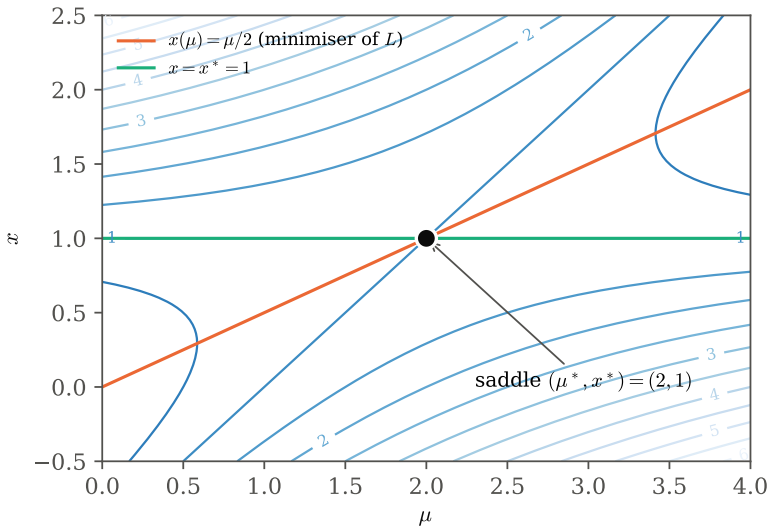


Figure 5: Contour plot of $\mathcal{L}(\mu, x)$. The line of minimum points $x(\mu) = \mu/2$ meets the line $x = x^*$ exactly at the saddle point $(2, 1)$.

2.2 (b) Why the saddle point is optimal

For any feasible x (that is, $h(x) \leq 0$),

$$f(x^*) \overset{(1)}{=} \mathcal{L}(\mu^*, x^*) \overset{(2)}{\leq} \mathcal{L}(\mu^*, x) \overset{(3)}{=} f(x) + \mu^* h(x) \overset{(4)}{\leq} f(x).$$

(1) $\mathcal{L}(\mu^*, x^*) = f(x^*) + \mu^* h(x^*)$, and $\mu^* h(x^*) = 0$ because $h(x^*) = 0$ (complementary slackness).

(2) This is the right inequality of the saddle point. x^* gives the minimum of $\mathcal{L}(\mu^*, \cdot)$ over *all* x, so it is also the minimum over the feasible x.

(3) This is the definition of the Lagrangian.

(4) $\mu^* \geq 0$, and $h(x) \leq 0$ for a feasible x. So $\mu^* h(x) \leq 0$.

Therefore $f(x^*) \leq f(x)$ for every feasible x, and x^* is the global minimum. Convexity is not used in this proof. So *any* saddle point of the Lagrangian gives an optimal solution of the primal problem.

What the left inequality says about $h(x^*)$. The left inequality is

$$f(x^*) + \mu\, h(x^*) \leq \mathcal{L}(\mu^*, x^*) < \infty \qquad \forall\, \mu \geq 0.$$

If $h(x^*) > 0$, the left side goes to $+\infty$ when $\mu \to \infty$. But the right side is a finite number, so this is not possible. Therefore $h(x^*) \leq 0$. This means that *the left inequality makes sure that x^* is feasible.* This is the same idea as the primal function g in Q1(c).

What it says about $\mu^* h(x^*)$. If $\mu = 0$ is used in the left inequality, $f(x^*) \leq f(x^*) + \mu^* h(x^*)$, so $\mu^* h(x^*) \geq 0$. Also, $\mu^* \geq 0$ and $h(x^*) \leq 0$, so $\mu^* h(x^*) \leq 0$. From these two results, $\mu^* h(x^*) = 0$. So the left inequality also gives the complementary slackness condition. It also means that μ^* gives the maximum of $\mathcal{L}(\cdot, x^*)$ over $\mu \geq 0$.

2.3 (c) Complementary slackness and an inactive constraint

Original problem. $\mu^* h(x^*) = 2\,(1 - 1) = 0$, so the complementary slackness condition is satisfied. In this case the constraint is active ($h(x^*) = 0$) and the multiplier is positive. The constraint actually stops the solution from going to the unconstrained minimum $x = 0$.

New constraint $x \geq -1$, that is, $h(x) = -1 - x \leq 0$.

- *Is $x = 0$ feasible?* Yes. $0 \geq -1$, or $h(0) = -1 \leq 0$. So the unconstrained minimum of x^2 is also the constrained minimum: $x^* = 0$ and $p^* = 0$.
- *Is the constraint active?* No. $h(0) = -1 < 0$, so the constraint is satisfied with a strict inequality at the optimum.
- *Expected value of μ^*: $\mu^* = 0$.* From complementary slackness, $\mu^* h(x^*) = \mu^* \cdot (-1) = 0$, so $\mu^* = 0$. The stationarity condition gives the same answer: $\partial_x \mathcal{L} = 2x^* - \mu^* = 0$ at $x^* = 0$, so $\mu^* = 0$. Also, if the bound -1 is changed by a small amount, the optimal cost does not change (p^* is still 0). So the shadow price is zero. Since an inactive constraint does not affect the solution, its multiplier is zero.

What the left inequality says about μ^*. With $x^* = 0$, $\mathcal{L}(\mu, x^*) = 0 + \mu(-1 - 0) = -\mu$. So the left inequality becomes $-\mu \leq -\mu^*$ for all $\mu \geq 0$, that is, $\mu^* \leq \mu$ for every $\mu \geq 0$. When $\mu = 0$, this gives $\mu^* \leq 0$. Since $\mu^* \geq 0$ as well, $\mu^* = 0$. So in this case, the left inequality gives only one possible value for the multiplier. In the active case, the left inequality was satisfied with equality for every μ.

The dual function gives the same result. Again $x(\mu) = \mu/2$, and $d(\mu) = \mu^2/4 + \mu(-1-\mu/2) = -\mu - \mu^2/4$. This function decreases for all $\mu \geq 0$, so $\mu^* = 0$ and $d^* = 0 = p^*$ (Figure 6). The numerical values (squares) are the same as the formula.

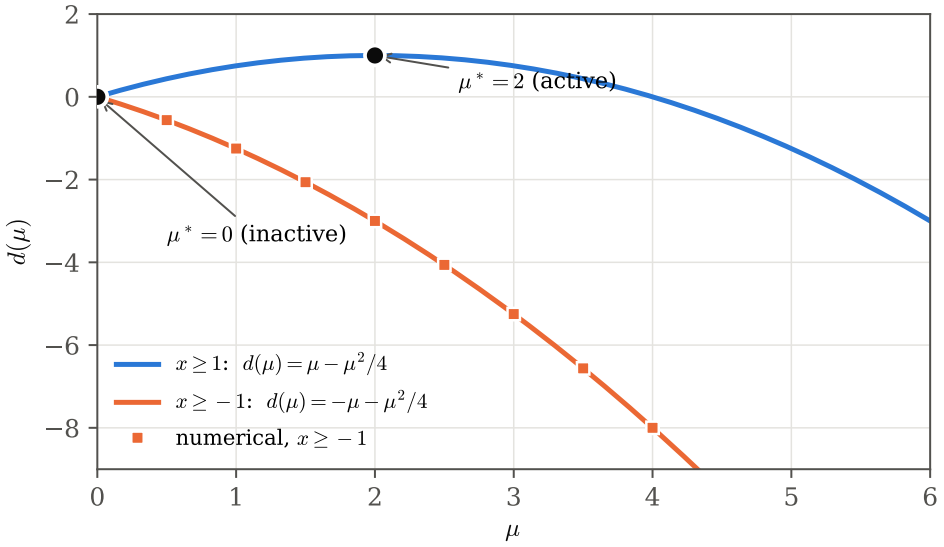


Figure 6: Dual functions for the active constraint $x \geq 1$ (maximum at $\mu^* = 2$) and for the inactive constraint $x \geq -1$ (maximum at $\mu^* = 0$).

3 Weak duality

3.1 (a) $d(\mu) \leq p^*$

Let $\mu \geq 0$, and let x be any feasible point, so $h(x) \leq 0$. Then

$$d(\mu) \overset{(1)}{\leq} \mathcal{L}(\mu, x) = f(x) + \mu h(x) \overset{(2)}{\leq} f(x).$$

(1) $d(\mu)$ is the minimum of $\mathcal{L}(\mu, \cdot)$ over *all* x. So it cannot be larger than the value of $\mathcal{L}(\mu, \cdot)$ at any one point x.

(2) $\mu \geq 0$ and $h(x) \leq 0$, so $\mu h(x) \leq 0$.

The result $d(\mu) \leq f(x)$ is true for *every* feasible x. So it is also true for the smallest value of $f(x)$. Taking the minimum over all feasible x,

$$d(\mu) \leq \min_{h(x) \leq 0} f(x) = p^* \qquad \forall\, \mu \geq 0.$$

Convexity is not used here, so weak duality is true for every problem.

Numerical check for Question 1, using $d(\mu) = \mu - \mu^2/4$ and $p^* = 1$:

In general, $p^* - d(\mu) = 1 - \mu + \mu^2/4 = (1 - \mu/2)^2 \geq 0$. So the result is true for every μ, not only for these four values. The simulation also shows that $\max_{\mu \in [0,100]} d(\mu) = 1 \leq p^*$. This means that any value of $d(\mu)$ gives a lower bound for the optimal cost, even without solving the primal problem.

3.2 (b) Lower envelope and concavity of d

When x is fixed, $\mathcal{L}(\mu, x) = x^2 + \mu(1 - x)$ is a straight line in μ. Its intercept is x^2 and its slope is $1 - x$:

$$x = 0 : \ \mathcal{L} = \mu, \qquad x = 1 : \ \mathcal{L} = 1, \qquad x = 2 : \ \mathcal{L} = 4 - \mu.$$

Table 3: Weak duality check. All the dual values are less than or equal to $p^* = 1$, and the gap becomes zero at $\mu^* = 2$.

μ	$d(\mu) = \mu - \mu^2/4$	p^*	gap $p^* - d(\mu)$	$d(\mu) \le p^*$?
0	0.00	1	1.00	✓
1	0.75	1	0.25	✓
2	1.00	1	0.00	✓ (equal)
3	0.75	1	0.25	✓

The lower envelope of these three lines is

$$\min\{\mu,\ 1,\ 4-\mu\} = \begin{cases} \mu, & 0 \le \mu \le 1, \\ 1, & 1 \le \mu \le 3, \\ 4-\mu, & \mu \ge 3. \end{cases}$$

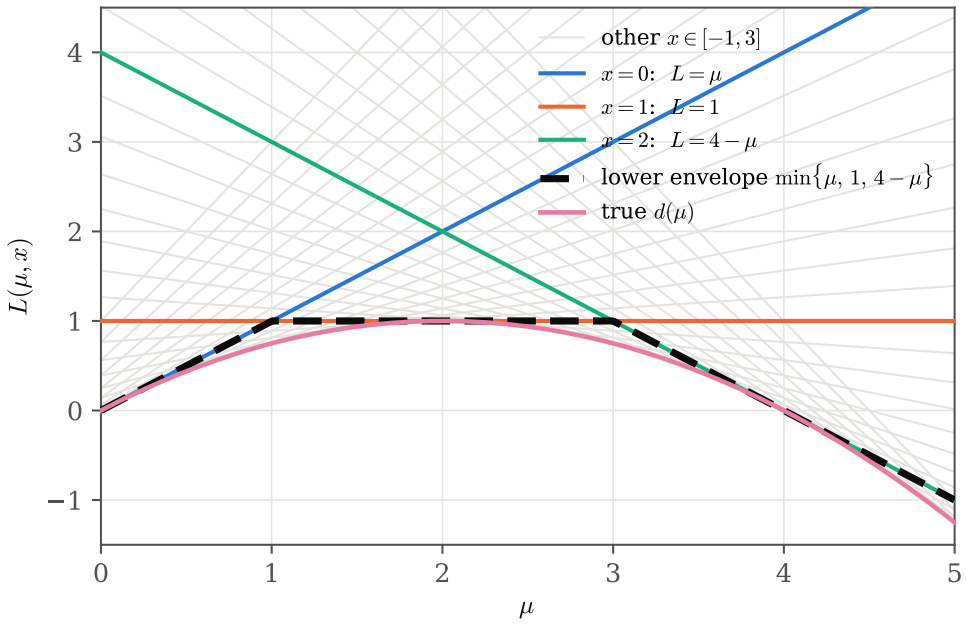


Figure 7: $\mathcal{L}(\mu, x)$ against μ for $x = 0, 1, 2$ (coloured lines) and for 33 other values of $x \in [-1, 3]$ (light grey). The dashed line is the lower envelope of the three coloured lines. It is made of straight pieces, it is concave, and it lies above $d(\mu)$. When more lines are used, the envelope comes closer to the true $d(\mu)$ (pink).

Concave or convex? The envelope is **concave**. Its slopes are $+1$, 0 and -1, so the slope never increases. The envelope is above $d(\mu)$, because it takes the minimum over only three values of x and not over all x. It touches $d(\mu)$ at $\mu = 0, 2, 4$, because at these points $x(\mu) = \mu/2$ is one of the chosen values $0, 1, 2$. When more lines are added (grey), the envelope becomes closer to $d(\mu)$.

Why d is concave for any f. For each fixed x, $f(x) + \mu h(x)$ is a straight line in μ, for any f and h. The dual function $d(\mu)$ is the minimum of all these lines (the lower envelope). The minimum of a set of straight lines is always concave. This can be shown as follows. For $\theta \in [0, 1]$,

$$d(\theta\mu_1 + (1-\theta)\mu_2) = \min_x \left[\theta\mathcal{L}(\mu_1, x) + (1-\theta)\mathcal{L}(\mu_2, x)\right] \ge \theta\, d(\mu_1) + (1-\theta)\, d(\mu_2).$$

Convexity of f is not needed for this. Therefore the dual problem is always a concave maximisation

problem, which is easy to solve, even when the primal problem is not convex. However, when the problem is not convex, strong duality may not hold, and there can be a gap $p^* - d^* > 0$.

4 Discussion and conclusions

- When the multiplier term is added, the minimum point of the Lagrangian, $x(\mu) = \mu/2$, moves towards the constraint boundary and then past it as μ increases. When $\mu^* = 2$, it is exactly at the constrained optimum $x^* = 1$.
- The dual function $d(\mu) = \mu - \mu^2/4$ is concave, and it is a lower bound for p^* for every $\mu \geq 0$ (weak duality). Its maximum $d^* = 1$ is equal to $p^* = 1$. So there is no duality gap, as expected from Slater's condition for this convex problem.
- The primal function $g(x) = \max_{\mu \geq 0} \mathcal{L}$ is equal to f in the feasible region and $+\infty$ outside it. So minimising g is the same as solving the original constrained problem.
- $(x^*, \mu^*) = (1, 2)$ is a saddle point of $\mathcal{L}$. The right inequality shows that x^* is optimal. The left inequality makes sure that x^* is feasible and that complementary slackness is satisfied.
- The multiplier shows how much the constraint affects the solution. For the active constraint $x \geq 1$, $\mu^* = 2 = \mathrm{d}p^*/\mathrm{d}b$. For the inactive constraint $x \geq -1$, $\mu^* = 0$.
- All the analytical results were checked using the simulation, and the numerical results agree with them.

A Numerical output of the simulation

The output below is given by running `src/duality_simulation.py`. It is also saved to `results/numerical_checks.txt`.

```
Q1(a): minimisers x(mu) = mu/2
  mu=0: x(mu) analytic=0.0000 numeric=0.0000 min L analytic=0.0000 numeric=0.0000
  mu=1: x(mu) analytic=0.5000 numeric=0.5000 min L analytic=0.7500 numeric=0.7500
  mu=2: x(mu) analytic=1.0000 numeric=1.0000 min L analytic=1.0000 numeric=1.0000
  mu=3: x(mu) analytic=1.5000 numeric=1.5000 min L analytic=0.7500 numeric=0.7500

Q1(b): mu* (grid search) = 2.0000, d* = 1.0000

Q1(c): truncated sup over mu in [0,M] at sample x
   x M=1 M=10 M=100 M=1000 x^2
  -1.00 3.000 21.000 201.000 2001.000 1.000
   0.00 1.000 10.000 100.000 1000.000 0.000
   0.50 0.750 5.250 50.250 500.250 0.250
   0.99 0.990 1.080 1.980 10.980 0.980
   1.00 1.000 1.000 1.000 1.000 1.000
   2.00 4.000 4.000 4.000 4.000 4.000
   3.00 9.000 9.000 9.000 9.000 9.000

Q2(a): L(mu*,x*) = 1.0000
  max_mu L(mu,x*) over mu in [0,50] = 1.0000 (left ineq holds: True)
  min_x L(mu*,x) over x in [-20,20] = 1.0000 (right ineq holds: True)
  max |L(2,x) - ((x-1)^2+1)| = 1.14e-13

Q2(c): x >= -1 -> argmin x^2 numeric = 0.0000, h(0) = -1.0
  mu* (grid search) = 0.0000, d* = 0.0000

Q3(a): weak duality check d(mu) <= p* = 1
  mu=0: d(mu) = 0.0000 gap p*-d = 1.0000 holds: True
  mu=1: d(mu) = 0.7500 gap p*-d = 0.2500 holds: True
  mu=2: d(mu) = 1.0000 gap p*-d = 0.0000 holds: True
  mu=3: d(mu) = 0.7500 gap p*-d = 0.2500 holds: True
  max over mu in [0,100] of d(mu) = 1.0000 <= 1

Q3(b): envelope min(mu, 1, 4-mu) >= d(mu) everywhere: True; touches d at mu = 0, 2, 4
  envelope midpoint-concave on grid: True
```

B Simulation source code

Repository: https://github.com/rajinthanr/en3330-lagrangian-duality

```python
"""
EN3330 Assignment 03 - Lagrangian duality for min x^2 s.t. x >= 1.

Generates every figure used in the report (saved as vector PDFs in ../figures)
and prints numerical checks of the analytical results (saved in ../results).

Run: python3 src/duality_simulation.py
Requires: numpy, matplotlib
"""

from pathlib import Path

import numpy as np
import matplotlib

matplotlib.use("Agg")
import matplotlib.pyplot as plt

ROOT = Path(__file__).resolve().parents[1]
FIG_DIR = ROOT / "figures"
RES_DIR = ROOT / "results"
FIG_DIR.mkdir(exist_ok=True)
RES_DIR.mkdir(exist_ok=True)

# ---------------------------------------------------------------- style ----
SERIES = ["#2a78d6", "#eb6834", "#1baf7a", "#eda100", "#e87ba4", "#008300"]
INK = "#0b0b0b"
INK_2 = "#52514e"
MUTED = "#8a8984"
GRID = "#e4e3df"
FEASIBLE = "#1baf7a"
INFEASIBLE = "#e34948"

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    "mathtext.fontset": "cm",
    "font.size": 10,
    "axes.edgecolor": INK_2,
    "axes.labelcolor": INK,
    "axes.titlesize": 10.5,
    "axes.grid": True,
    "grid.color": GRID,
    "grid.linewidth": 0.6,
    "xtick.color": INK_2,
    "ytick.color": INK_2,
    "lines.linewidth": 2.0,
    "legend.frameon": False,
    "legend.fontsize": 9,
    "savefig.bbox": "tight",
})


def save(fig, name):
    fig.savefig(FIG_DIR / f"{name}.pdf")
    fig.savefig(FIG_DIR / f"{name}.png", dpi=200)
    plt.close(fig)


# ---------------------------------------------------------------- problem ----
def f(x):
    """Objective f(x) = x^2."""
    return x ** 2

```

```python

def h(x, b=1.0):
    """Constraint h(x) = b - x <= 0 (i.e. x >= b)."""
    return b - x


def L(mu, x, b=1.0):
    """Lagrangian L(mu, x) = f(x) + mu h(x)."""
    return f(x) + mu * h(x, b)


def x_of_mu(mu):
    """Analytical minimiser of L(mu, .): dL/dx = 2x - mu = 0."""
    return mu / 2.0


def d_analytic(mu):
    """Dual function d(mu) = mu - mu^2/4."""
    return mu - mu ** 2 / 4.0


X_GRID = np.linspace(-10.0, 10.0, 400_001) # fine grid for numerical minimisation


def d_numeric(mu, b=1.0):
    vals = L(mu, X_GRID, b)
    i = np.argmin(vals)
    return vals[i], X_GRID[i]


log_lines = []


def log(s=""):
    print(s)
    log_lines.append(s)


# ============================================================================ Q1(a) ==
mus = [0, 1, 2, 3]
x = np.linspace(-1.5, 3.0, 600)
fig, ax = plt.subplots(figsize=(6.0, 3.9))
ax.axvspan(-1.5, 1.0, color=INFEASIBLE, alpha=0.06, lw=0)
ax.axvline(1.0, color=MUTED, lw=1.0, ls="--")
ax.text(0.97, 9.3, "infeasible $x<1$", ha="right", color=INK_2, fontsize=8.5)
ax.text(1.05, 9.3, "feasible $x\\geq 1$", ha="left", color=INK_2, fontsize=8.5)
for k, mu in enumerate(mus):
    c = SERIES[k]
    ax.plot(x, L(mu, x), color=c, label=f"$\\mu={mu}$")
    xm = x_of_mu(mu)
    ax.plot(xm, L(mu, xm), "o", ms=7, color=c, mec="white", mew=1.5, zorder=5)
    off = [(0, -15), (-16, -15), (8, -17), (16, -15)][k]
    ax.annotate(f"$x({mu})={xm:g}$", (xm, L(mu, xm)), textcoords="offset points",
                xytext=off, ha="center", fontsize=8.5, color=INK)
# trajectory of minimisers: (mu/2, mu - mu^2/4) lies on the curve y = 2x - x^2
mt = np.linspace(0, 3, 100)
ax.plot(x_of_mu(mt), d_analytic(mt), color=INK_2, lw=1.0, ls=":",
        label="path of $x(\\mu)$")
ax.set_xlim(-1.5, 3.0)
ax.set_ylim(-1.0, 10.0)
ax.set_xlabel("$x$")
ax.set_ylabel("$L(\\mu,x)=x^2+\\mu(1-x)$")
ax.legend(loc="upper left", ncol=1)
save(fig, "q1a_lagrangian_vs_x")

log("Q1(a): minimisers x(mu) = mu/2")
for mu in mus:
```

```
    Ln, xn = d_numeric(mu)
    log(f" mu={mu}: x(mu) analytic={x_of_mu(mu):.4f} numeric={xn:.4f} "
        f"min L analytic={d_analytic(mu):.4f} numeric={Ln:.4f}")

# ================================================================ Q1(b) ==
mu = np.linspace(0, 6, 400)
fig, ax = plt.subplots(figsize=(6.0, 3.5))
ax.axhline(1.0, color=MUTED, lw=1.0, ls="--")
ax.text(5.95, 1.07, "$p^*=f(x^*)=1$", ha="right", color=INK_2, fontsize=8.5)
ax.plot(mu, d_analytic(mu), color=SERIES[0], label="$d(\\mu)=\\mu-\\mu^2/4$ (analytic)")
mu_s = np.arange(0, 6.01, 0.5)
ax.plot(mu_s, [d_numeric(m)[0] for m in mu_s], "s", ms=5, color=SERIES[1],
        mec="white", mew=1.0, label="numerical minimum of $L(\\mu,\\cdot)$ on a grid")
ax.plot(2, 1, "o", ms=9, color=INK, mec="white", mew=1.5, zorder=6)
ax.annotate("$\\mu^*=2,\\ d^*=1$", (2, 1), xytext=(2.5, 1.3),
            arrowprops=dict(arrowstyle="->", color=INK_2, lw=0.8), fontsize=9)
ax.set_xlim(0, 6)
ax.set_ylim(-3.2, 1.6)
ax.set_xlabel("$\\mu$")
ax.set_ylabel("$d(\\mu)$")
ax.legend(loc="lower left")
save(fig, "q1b_dual_function")

mu_dense = np.linspace(0, 10, 100_001)
i = np.argmax(d_analytic(mu_dense))
log(f"\nQ1(b): mu* (grid search) = {mu_dense[i]:.4f}, d* = {d_analytic(mu_dense)[i]:.4f}")

# ================================================================ Q1(c) ==
# g(x) = sup_{mu>=0} L(mu,x). Approximate the sup by mu in [0, M] for growing M.
xg = np.linspace(-1, 3, 801)
fig, ax = plt.subplots(figsize=(6.0, 3.9))
ax.axvspan(-1, 1, color=INFEASIBLE, alpha=0.06, lw=0)
ax.axvline(1.0, color=MUTED, lw=1.0, ls="--")
for k, M in enumerate([0, 2, 5, 10, 20]):
    gM = np.max([L(m, xg) for m in np.linspace(0, M, 201)], axis=0)
    ax.plot(xg, gM, color=SERIES[k], lw=1.4, alpha=0.9, label=f"$M={M}$")
# exact primal function
xf = xg[xg >= 1]
ax.plot(xf, f(xf), color=INK, lw=2.6, label="$g(x)=x^2$ for $x\\geq1$")
ax.plot(1, 1, "o", ms=8, color=INK, mec="white", mew=1.5, zorder=6)
ax.annotate("$\\min g = g(1)=1$", (1, 1), xytext=(1.6, 0.0),
            arrowprops=dict(arrowstyle="->", color=INK_2, lw=0.8), fontsize=9)
for xa in [-0.75, -0.25, 0.25, 0.75]:
    ax.annotate("", xy=(xa, 21.5), xytext=(xa, 15.5),
                arrowprops=dict(arrowstyle="-|>", color=INFEASIBLE, lw=1.2))
ax.text(0.0, 13.4, "$g(x)=+\\infty$ for $x<1$", ha="center", color=INFEASIBLE, fontsize=9.5,
        bbox=dict(boxstyle="round,pad=0.25", fc="white", ec="none"), zorder=7)
ax.set_xlim(-1, 3)
ax.set_ylim(-1.5, 22)
ax.set_xlabel("$x$")
ax.set_ylabel("$g(x)$")
ax.legend(loc="upper right", fontsize=8,
          title="$\\sup$ of $L(\\mu,x)$ over $\\mu\\in[0,M]$", title_fontsize=8)
save(fig, "q1c_primal_function")

log("\nQ1(c): truncated sup over mu in [0,M] at sample x")
log(" x " + "".join(f"M={M:<9}" for M in [1, 10, 100, 1000]) + "x^2")
for xv in [-1.0, 0.0, 0.5, 0.99, 1.0, 2.0, 3.0]:
    row = "".join(f"{max(L(0, xv), L(M, xv)):<11.3f}" for M in [1, 10, 100, 1000])
    log(f" {xv:5.2f} {row}{f(xv):.3f}")

# ================================================================ Q2(a) ==
x_star, mu_star = 1.0, 2.0
fig, axes = plt.subplots(1, 2, figsize=(6.4, 2.8))
ax = axes[0]
mm = np.linspace(0, 6, 100)
ax.plot(mm, L(mm, x_star), color=SERIES[0])
```

```
ax.plot(mu_star, L(mu_star, x_star), "o", ms=8, color=INK, mec="white", mew=1.5, zorder=5)
ax.annotate("$L(\\mu^*,x^*)=1$", (mu_star, 1), xytext=(2.5, 1.6),
          arrowprops=dict(arrowstyle="->", color=INK_2, lw=0.8), fontsize=9)
ax.set_xlim(0, 6)
ax.set_ylim(0, 3)
ax.set_xlabel("$\\mu$")
ax.set_ylabel("$L(\\mu,x^*)$")
ax.set_title("$L(\\mu,1)=1$ for all $\\mu\\geq 0$")
ax = axes[1]
xx = np.linspace(-1, 3, 300)
ax.plot(xx, L(mu_star, xx), color=SERIES[1])
ax.plot(x_star, L(mu_star, x_star), "o", ms=8, color=INK, mec="white", mew=1.5, zorder=5)
ax.axhline(1, color=MUTED, lw=1.0, ls="--")
ax.annotate("$L(\\mu^*,x^*)=1$", (1, 1), xytext=(1.5, 3.0),
          arrowprops=dict(arrowstyle="->", color=INK_2, lw=0.8), fontsize=9)
ax.set_xlim(-1, 3)
ax.set_ylim(0, 5.5)
ax.set_xlabel("$x$")
ax.set_ylabel("$L(\\mu^*,x)$")
ax.set_title("$L(2,x)=(x-1)^2+1$")
fig.tight_layout(w_pad=2.0)
save(fig, "q2a_saddle_slices")

# 3-D style contour view of the saddle point
MU, XX = np.meshgrid(np.linspace(0, 4, 200), np.linspace(-0.5, 2.5, 200))
fig, ax = plt.subplots(figsize=(5.2, 3.6))
cs = ax.contour(MU, XX, L(MU, XX), levels=np.arange(-2, 7, 0.5), cmap="Blues_r", linewidths=1.0)
ax.clabel(cs, cs.levels[::2], fontsize=7, fmt="%g")
ax.plot(mt := np.linspace(0, 4, 50), x_of_mu(mt), color=SERIES[1], lw=1.4,
       label="$x(\\mu)=\\mu/2$ (minimiser of $L$)")
ax.axhline(1, color=SERIES[2], lw=1.4, label="$x=x^*=1$")
ax.plot(mu_star, x_star, "o", ms=9, color=INK, mec="white", mew=1.5, zorder=6)
ax.annotate("saddle $(\\mu^*,x^*)=(2,1)$", (2, 1), xytext=(2.3, 0.0),
          arrowprops=dict(arrowstyle="->", color=INK_2, lw=0.8), fontsize=9)
ax.set_xlabel("$\\mu$")
ax.set_ylabel("$x$")
ax.legend(loc="upper left", fontsize=8)
ax.grid(False)
save(fig, "q2a_saddle_contour")

# numerical saddle check on a grid
mu_chk = np.linspace(0, 50, 5001)
x_chk = np.linspace(-20, 20, 40001)
left_ok = np.all(L(mu_chk, x_star) <= L(mu_star, x_star) + 1e-12)
right_ok = np.all(L(mu_star, x_star) <= L(mu_star, x_chk) + 1e-12)
log(f"\nQ2(a): L(mu*,x*) = {L(mu_star, x_star):.4f}")
log(f" max_mu L(mu,x*) over mu in [0,50] = {L(mu_chk, x_star).max():.4f} (left ineq holds: {left_ok})")
log(f" min_x L(mu*,x) over x in [-20,20] = {L(mu_star, x_chk).min():.4f} (right ineq holds: {right_ok})")
log(f" max |L(2,x) - ((x-1)^2+1)| = {np.abs(L(2, x_chk) - ((x_chk - 1) ** 2 + 1)).max():.2e}")

# ============================================================================ Q2(c) ==
# modified constraint x >= -1 : h(x) = -1 - x
b2 = -1.0
mu = np.linspace(0, 6, 400)
fig, ax = plt.subplots(figsize=(6.0, 3.3))
ax.plot(mu, d_analytic(mu), color=SERIES[0], label="$x\\geq 1$: $d(\\mu)=\\mu-\\mu^2/4$")
ax.plot(mu, -mu - mu ** 2 / 4, color=SERIES[1], label="$x\\geq -1$: $d(\\mu)=-\\mu-\\mu^2/4$")
ax.plot(mu_s, [d_numeric(m, b2)[0] for m in mu_s], "s", ms=5, color=SERIES[1],
       mec="white", mew=1.0, label="numerical, $x\\geq -1$")
ax.plot(2, 1, "o", ms=8, color=INK, mec="white", mew=1.5, zorder=6)
ax.plot(0, 0, "o", ms=8, color=INK, mec="white", mew=1.5, zorder=6)
ax.annotate("$\\mu^*=2$ (active)", (2, 1), xytext=(2.6, 0.2),
          arrowprops=dict(arrowstyle="->", color=INK_2, lw=0.8), fontsize=9)
ax.annotate("$\\mu^*=0$ (inactive)", (0, 0), xytext=(0.5, -3.5),
          arrowprops=dict(arrowstyle="->", color=INK_2, lw=0.8), fontsize=9)
ax.set_xlim(0, 6)
ax.set_ylim(-9, 2)
```

```
ax.set_xlabel("$\\mu$")
ax.set_ylabel("$d(\\mu)$")
ax.legend(loc="lower left", fontsize=8.5)
save(fig, "q2c_modified_dual")

mu_dense = np.linspace(0, 10, 100_001)
d2 = np.array([-m - m ** 2 / 4 for m in mu_dense])
log(f"\nQ2(c): x >= -1 -> argmin x^2 numeric = {d_numeric(0, b2)[1]:.4f}, h(0) = {h(0, b2):.1f}")
log(f" mu* (grid search) = {mu_dense[np.argmax(d2)]:.4f}, d* = {d2.max() + 0.0:.4f}")

# ================================================================== Q3(a) ==
log("\nQ3(a): weak duality check d(mu) <= p* = 1")
for m in [0, 1, 2, 3]:
    log(f" mu={m}: d(mu) = {d_analytic(m):.4f} gap p*-d = {1 - d_analytic(m):.4f} "
        f"holds: {d_analytic(m) <= 1}")
mu_dense = np.linspace(0, 100, 100_001)
log(f" max over mu in [0,100] of d(mu) = {d_analytic(mu_dense).max():.4f} <= 1")

# ================================================================== Q3(b) ==
mu = np.linspace(0, 5, 400)
fig, ax = plt.subplots(figsize=(6.0, 3.8))
for xv in np.linspace(-1, 3, 33): # many faint lines -> envelope converges to d
    ax.plot(mu, L(mu, xv), color=GRID, lw=0.8, zorder=1,
            label="other $x\\in[-1,3]$" if xv == -1 else None)
for k, xv in enumerate([0, 1, 2]):
    ax.plot(mu, L(mu, xv), color=SERIES[k], lw=1.6, zorder=3,
            label=f"$x={xv}$: $L={['\\mu', '1', '4-\\mu'][k]}$")
env = np.min([L(mu, xv) for xv in [0, 1, 2]], axis=0)
ax.plot(mu, env, color=INK, lw=2.8, ls=(0, (4, 2)), zorder=4,
        label="lower envelope $\\min\\{\\mu,\\,1,\\,4-\\mu\\}$")
ax.plot(mu, d_analytic(mu), color=SERIES[4], lw=1.8, zorder=5, label="true $d(\\mu)$")
ax.set_xlim(0, 5)
ax.set_ylim(-1.5, 4.5)
ax.set_xlabel("$\\mu$")
ax.set_ylabel("$L(\\mu,x)$")
ax.legend(loc="upper right", fontsize=8.5)
save(fig, "q3b_lower_envelope")

env_check = np.min([L(mu, xv) for xv in [0, 1, 2]], axis=0)
log("\nQ3(b): envelope min(mu, 1, 4-mu) >= d(mu) everywhere: "
    f"{np.all(env_check >= d_analytic(mu) - 1e-12)}; touches d at mu = 0, 2, 4")
# midpoint concavity test of the envelope
a, c = np.meshgrid(mu, mu)
envf = lambda m: np.minimum(np.minimum(m, 1.0), 4.0 - m)
log(" envelope midpoint-concave on grid: "
    f"{np.all(envf((a + c) / 2) >= (envf(a) + envf(c)) / 2 - 1e-12)}")

(RES_DIR / "numerical_checks.txt").write_text("\n".join(log_lines) + "\n")
print(f"\nFigures written to {FIG_DIR}\nLog written to {RES_DIR / 'numerical_checks.txt'}")
```